

Guardian AI Creative Studio — System Design Document

Core Structure

The Guardian AI Creative Studio is an integrated AI-powered platform combining design, generation, and performance analytics. It enables users to create, optimize, preview, and export marketing creatives with AI assistance. The system operates through three main modes:

1. **Image Generation**
2. **Video Generation**
3. **Caption Generation**

Each mode is accessible via dynamic tabs that update the interface accordingly.

1. Image Generation Mode

Fields

- **Prompt Input (large textarea)**
Placeholder: “ Describe the image you want to generate...”
- **Style Selection**
 - Modern
 - Minimal
 - Bold
 - Playful
 - Corporate
 - Vintage
- **Aspect Ratio**
 - 1:1 (Instagram)
 - 4:5
 - 9:16 (Reels/Stories)
 - 16:9 (YouTube)
 - 3:4
- **Platform Presets**
 - Instagram Post
 - Instagram Story
 - YouTube Thumbnail
 - LinkedIn Ad
 - Banner Ad
- **Brand Settings**
 - Upload Logo
 - Brand Colors
 - Font Style

Buttons

- Generate Image
 - Export All
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2. Video Generation Mode

Fields

- **Video Prompt**
- **Duration**
 - 5 sec
 - 10 sec
 - 30 sec
- **Style**
 - Ad
 - Cinematic
 - Social Media
- **Voiceover**
 - Male
 - Female
 - None
- **Background Music Toggle**

Output

- AI-generated video preview
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3. Caption Generation Mode

Fields

- **Caption Prompt / Product Description**
- **Tone**
 - Professional
 - Casual
 - Emotional
 - Sales-driven
- **Platform**
 - Instagram
 - LinkedIn
 - Twitter

Output

- Multiple caption variations

- Hashtags
 - CTA suggestions
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4. Generated Output Section

Generated creatives are displayed in a card-based preview system.

Each Card Includes

- Preview (image/video/text)
- Title (e.g., “AI Generated Image”)
- Style + Aspect Ratio
- Metrics:
 - Quality Score (%)
 - Predicted CTR (%)
 - Engagement Level

Buttons

- Download
 - Edit
 - Regenerate
 - Send to Campaign
 - Run Ads
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5. AI Performance Intelligence

Each creative is analyzed for performance potential.

Metrics

- Predicted CTR
- Engagement Score
- Conversion Probability

Analysis Factors

- Visual appeal
 - Copy strength
 - Platform suitability
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6. AI Creative Enhancer

Smart AI actions to improve creative performance.

Actions

- Improve Quality
 - Optimize for CTR
 - Add Branding (logo/colors)
 - Resize for platforms
 - Generate variations
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7. Compliance & Safety Check

Automated scanning for policy and content compliance.

Checks

- Policy violations
- Misleading claims
- Sensitive content

Output

- Risk Level: Low / Medium / High
 - Fix Suggestions
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8. Creative Library

A dedicated section for managing and retrieving creatives.

Features

- Saved creatives
 - History
 - Filters:
 - Campaign
 - Platform
 - Date
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9. Integration with System

Seamless connectivity across Guardian AI modules.

Integrations

- **Campaign Dashboard** → Send to Campaign
 - **AI Assistant** → Generate via chat
 - **Ads Engine** → Run Ads directly
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10. Real-Time Preview

Dynamic previews showing how creatives appear on major platforms.

Supported Platforms

- Instagram
 - Facebook
 - LinkedIn
 - YouTube
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UI/UX Design

Design Principles

- Clean, modern interface
- Card-based layout
- Rounded corners
- Soft shadows
- Smooth transitions

Color Indicators

- Green → High performance
 - Yellow → Medium
 - Red → Risk
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Final Experience

The Guardian AI Creative Studio delivers an intelligent, end-to-end creative experience:

This system functions as a **Creative Intelligence Engine**, merging design, generation, and analytics into one unified platform.